	him.	
7(a)(i)	Energy density is the amount of energy that can be released per unit mass of fuel.	[1] for correct answer
7(a)(ii)	Rate of consumption of natural gas $= \dfrac{\dfrac{\text{total electrical power output}}{\text{efficiency of power stations}}}{\text{energy density of natural gas}}$ $= \dfrac{\dfrac{12600 \times 10^6}{0.27}}{56 \times 10^6}$ $= \dfrac{12600}{0.27 \times 56}$ $= 833$ $\approx 830 \text{ kg s}^{-1}$	[1] for correct method and numerical substitution [1] for correct answer
7(b)(i)	Solar intensity incident on Earth $I = \dfrac{\text{radiant power of the Sun, } P}{\text{area of perpendicular surface, } A}$ $= \dfrac{3.90 \times 10^{26}}{4\pi \times \left(1.50 \times 10^{11}\right)^2}$ $= 1379$ $\approx 1400 \text{ W m}^{-2}$	[1] for correct expression and numerical substitution [1] for correct answer
7(b)(ii)	The Sun's energy is radiated/emitted uniformly in all directions.	[1] for correct answer
7(b)(iii)	1. Cloudy times / cloud cover / sun is not directly overhead all the time 2. Absorption in atmosphere due to ozone, carbon dioxide and water vapour/humidity 3. Distance between Earth and Sun is not constant.	[1] for correct answer [1] for correct answer
7(c)(i)	Connecting many PV cells in series increases the e.m.f. of the electrical supply (allow higher voltage).	[1] for correct answer

No.	Solution	Remarks
7(c)(ii)	Currents from each parallel section of PV cells add up at the junction to the external circuit to provide a useful current supply. or The output current of the PV cell array is increased. If one PV cell is faulty the others still work and continue to supply energy. Connecting many PV cells in parallel lowers total resistance of the PV cell array, in the same principle that resistors in parallel result in an equivalent resistance that is always lower than every individual resistor.	[1] for correct answer [1] for correct answer
7(d)	$\dfrac{\text{solar power output from 37 km}^2\text{ of PV panels}}{\text{total power output from power stations}}$ $=\dfrac{780\times37\times10^6\times0.20}{12600\times10^6}$ $=0.4581$ ≈0.46	[1] for correct numerical substitution [1] for correct answer
7(e)(i)1.	Power produced by the PV panel $=$ average solar intensity $\times$ area of PV panel $\times$ efficiency of PV panel $=780\times1.50\times0.20$ $=234$ W	[1] for correct numerical substitution [1] for correct answer
7(e)(i)2.	Total number of sunshine hours in 2020 $=170+185+195+175+180+180$ $+190+180+155+155+130+135$ $=2030$ hours Electrical energy produced by the PV panel in 2020 $=\dfrac{234}{1000}\times2030$ $=475.02$ ≈475 kWh	[1] for correct numerical substitution [1] for correct answer
7(e)(ii)	Cost of saving in electricity per year $=$ electrical energy produced by PV power $\times$ electricity tariff $=475\times0.23$ $=\$109.25$ Payback period or no. of years to recover the cost of installing the PV panel	[1] for cost savings

No.	Solution	Remarks
	$= \dfrac{750}{109.25}$ $= 6.9$ years It takes about 7 years for the payback period for the investment of the PV panel. This is about a third of the useful lifespan of 25 years for the PV panel. Hence, the calculations are for the use of solar power, in addition to the benefit of reducing the intensity of greenhouse gas emissions.	[1] for payback period [1] for conclusion with valid explanation